EXTENDING A DUALIZING COMPLEX ALONG A POLYNOMIAL RING STAYS DUALIZING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The finiteness of the injective dimension of the extension of scalars of the dualizing complex. Invoked at the start of the counted proof (source lines 3075-3077). Establishes the first defining property of a dualizing complex for the polynomial ring.
- The base change behaviour of the derivation of the internal hom along a flat map. Invoked at the end of the counted proof (source lines 3084-3086). Turns the duality quasi-isomorphism over the source into the corresponding quasi-isomorphism over the polynomial ring.

Source reference. Original lemma lemma-injective-dimension-over-polynomial-ring: sources/stacks-dualizing/pinned-original-source.tex, source lines 591–591. Navigation pointer only: the counted text refers to the Stacks item lemma-injective-dimension-over-polynomial-ring, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-base-change-RHom: sources/stacks-dualizing/pinned-original-source.tex, source lines 2719–2719. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-base-change-RHom, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Source reference. Original reference more-algebra-lemma-finite-injective-dimension: sources/stacks-dualizing/pinned-original-source.tex, source lines 3076–3076. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-finite-injective-dimension, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let A be a Noetherian ring. If $\omega_A^\bullet$ is a dualizing complex, then $\omega_A^\bullet \otimes_A A[x]$ is a dualizing complex for $A[x]$.*

Proof. Set $B = A[x]$ and $\omega_B^\bullet = \omega_A^\bullet \otimes_A B$. It follows from Lemma lemma-injective-dimension-over-polynomial-ring and More on Algebra, Lemma more-algebra-lemma-finite-injective-dimension that $\omega_B^\bullet$ has finite injective dimension. Since $H^i(\omega_B^\bullet) = H^i(\omega_A^\bullet) \otimes_A B$ by flatness of $A \to B$ we see that $\omega_A^\bullet \otimes_A B$ has finite cohomology modules. Finally, the map

$$B \longrightarrow R\operatorname{Hom}_B(\omega_B^\bullet, \omega_B^\bullet)$$

is a quasi-isomorphism as formation of internal hom commutes with flat base change in this case, see More on Algebra, Lemma more-algebra-lemma-base-change-RHom. □